The Neural Collapse–Calibration Connection is Dataset-Dependent: An Empirical Investigation via Controlled Within-Class Geometry

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Abstract

Neural collapse (NC) — the phenomenon where deep network representations converge to a symmetric, maximally-separated structure during training — has been hypothesized to harm model calibration by destroying within-class geometric information needed for uncertainty estimation. We investigate this hypothesis through a systematic empirical study using the *Calibrated Collapse Regularizer* (CCR), a training-time method that maintains minimum within-class feature spread to prevent excessive NC1 collapse. Across CIFAR-10, CIFAR-100, and TinyImageNet with ResNet-18, we find that CCR successfully controls within-class spread (2–12$\times$ higher than standard training) with modest accuracy cost. However, our central finding is that the NC1–calibration relationship exhibits a *dataset-dependent sign flip*: on CIFAR-10, higher NC1 (less collapse) significantly correlates with lower ECE ($\rho = -0.65$, $p = 0.003$), consistent with the hypothesis. On CIFAR-100, the correlation reverses — higher NC1 correlates with *higher* ECE ($\rho = +0.68$, $p = 0.002$). On TinyImageNet, CCR reduces ECE by 36% but at a 16% accuracy cost. These results demonstrate that controlling neural collapse alone does not reliably improve calibration, and that the optimal representation geometry is fundamentally dataset-dependent. We provide detailed analysis of why this sign flip occurs and discuss implications for future representation-aware calibration methods.

1 Introduction

Deep neural networks trained with supervised objectives exhibit a striking geometric phenomenon in their final-layer representations known as *neural collapse* (NC) [Papyan et al., 2020]. As training progresses beyond zero training error, four interrelated properties emerge: (NC1) within-class feature variability collapses to zero; (NC2) class means converge to the vertices of a simplex equiangular tight frame (ETF); (NC3) the last-layer classifier converges to the class means (self-duality); and (NC4) classification reduces to nearest-class-center decisions. Neural collapse has been established as globally optimal under various theoretical settings [Zhu et al., 2021] and has become a central framework for understanding supervised representation learning.

Separately, *model calibration* — the alignment between predicted confidence and actual correctness probability — is critical for reliable deployment of deep learning systems [Guo et al., 2017]. Overconfident predictions can lead to failures in safety-critical applications such as medical diagnosis and autonomous driving. Modern neural networks are notoriously miscalibrated, typically exhibiting severe overconfidence [Guo et al., 2017].

A natural question arises: *does neural collapse contribute to miscalibration?* The intuition is compelling. As NC1 drives within-class features to collapse onto their class means, all samples —

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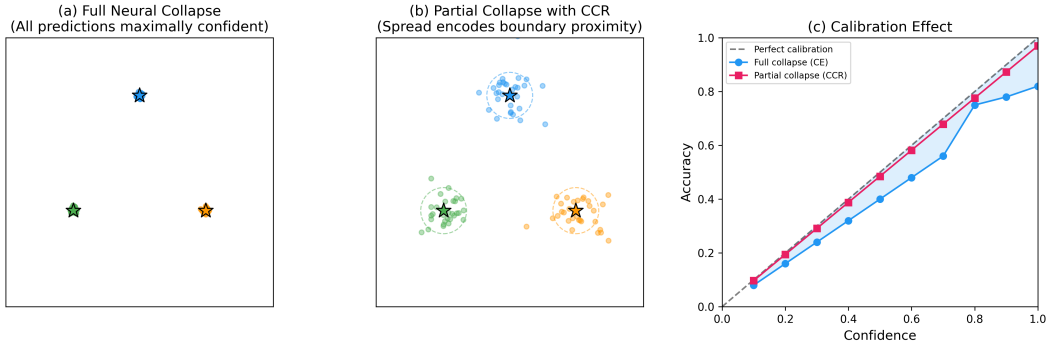

Figure 1: Conceptual illustration of our approach. **(a)** Full neural collapse: within-class features collapse to points, producing uniformly high-confidence predictions. **(b)** Partial collapse with CCR: features maintain spread that encodes proximity to decision boundaries. **(c)** The expected calibration effect: partial collapse should produce predictions closer to the identity line (perfect calibration).

regardless of proximity to decision boundaries — produce nearly identical, maximally confident predictions. This eliminates the geometric signal that could encode prediction uncertainty. Recent work has begun exploring this connection: Guo et al. [2025] analyzed how label smoothing affects calibration through the NC lens, and Harun et al. [2025] showed that controlling NC degree benefits out-of-distribution detection.

In this work, we directly test the hypothesis that *controlling the degree of NC1 collapse improves calibration*. We propose the Calibrated Collapse Regularizer (CCR), a training-time regularizer that maintains a minimum level of within-class feature spread through a hinge-style penalty (Figure 1).

Our central finding is a dataset-dependent sign flip in the NC1–calibration relationship. On CIFAR-10, the Spearman correlation between NC1 and ECE across methods and hyperparameter settings is $\rho = -0.65$ ($p = 0.003$), consistent with the hypothesis that less collapse improves calibration. On CIFAR-100, this relationship *reverses*: $\rho = +0.68$ ($p = 0.002$), meaning less collapse is associated with *worse* calibration. This sign flip is itself a significant finding that reveals the NC–calibration connection is more complex than any simple monotonic relationship.

Our contributions are:

- We propose CCR, a family of training-time regularizers that control within-class feature geometry by enforcing a minimum spread threshold, with fixed-threshold, adaptive-threshold, and softplus variants.
- We conduct a systematic empirical study across three datasets (CIFAR-10, CIFAR-100, TinyImageNet), measuring both NC metrics and calibration metrics for cross-entropy, label smoothing, Mixup, and CCR variants with 3 random seeds.
- We discover a *dataset-dependent sign flip* in the NC1–ECE correlation: negative on CIFAR-10 ($\rho = -0.65$), positive on CIFAR-100 ($\rho = +0.68$), demonstrating that no single NC-based prescription universally improves calibration.
- We document that label smoothing severely degrades raw calibration on CIFAR ($3\times$ higher ECE) but improves it on TinyImageNet ($2\times$ lower ECE), despite reducing within-class spread in both cases — further evidence of dataset-dependency.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the CCR regularizer. Section 4 presents experimental results. Section 5 analyzes the NC–calibration relationship and the sign flip phenomenon. Section 6 discusses limitations and implications.

2 Related Work

Neural Collapse. Neural collapse was identified by Papyan et al. [2020] as a terminal phase of training in deep classifiers, characterized by four interrelated geometric properties (NC1–NC4).

Zhu et al. [2021] proved NC is globally optimal for the unconstrained features model. Xu and Liu [2023] proposed the Variability Collapse Index to quantify NC1 degree. Most relevant to our work, Harun et al. [2025] proposed controlling the degree of NC using entropy regularization, finding that NC degree affects OOD detection and transfer learning differently. Our work extends this line of investigation to calibration.

Calibration Methods. Post-hoc calibration methods include temperature scaling [Guo et al., 2017], Platt scaling [Platt, 1999], and histogram binning [Zadrozny and Elkan, 2001]. Training-time methods include label smoothing [Szegedy et al., 2016, Müller et al., 2019] and Mixup [Zhang et al., 2018]. More recently, Tao et al. [2025] proposed Feature Clipping, a post-hoc method that clips feature magnitudes to reduce overconfidence, observing that high-ECE samples tend to have larger feature variance — an observation that partially motivated our work. Ni et al. [2025] proposed BalCAL, which balances a learnable classifier with a fixed ETF classifier for calibration.

NC–Calibration Connection. Guo et al. [2025] analyzed the relationship between neural collapse and calibration through the lens of label smoothing, finding that label smoothing accelerates NC and can cause overconfidence through excessive NC1 collapse. Park et al. [2024] analyzed how regularization affects calibration through the representation space. Our work differs from these analytical studies by proposing a method specifically designed to control NC1 for calibration, and by revealing the dataset-dependent nature of the NC–calibration relationship.

Supervised Representation Learning. Supervised contrastive learning [Khosla et al., 2020] and its variants [Wang et al., 2025] learn representations by pulling same-class features together and pushing different-class features apart. These methods implicitly control NC geometry. Our CCR regularizer is complementary and can be combined with any supervised training objective.

3 Method

3.1 Background: Neural Collapse and Calibration

Consider a classification model $f_\theta : \mathcal{X} \to \mathbb{R}^K$ that maps inputs to logits over K classes. Let $\phi_\theta(x) \in \mathbb{R}^d$ denote the penultimate-layer features. During the terminal phase of training, neural collapse manifests as: the within-class feature covariance $\Sigma_W = \frac{1}{N}\sum_{c=1}^{K}\sum_{i:y_i=c}(\phi_\theta(x_i) - \boldsymbol{\mu}_c)(\phi_\theta(x_i) - \boldsymbol{\mu}_c)^\top$ approaches zero (NC1), where $\boldsymbol{\mu}_c$ is the mean of class-c features.

The NC1 metric is commonly measured as $\text{NC1} = \frac{1}{K}\text{tr}(\Sigma_W \Sigma_B^{-1})$, where Σ_B is the between-class covariance. A more interpretable measure is the *within-class spread*: $S_c = \frac{1}{|\mathcal{D}_c|}\sum_{i\in\mathcal{D}_c}\|\phi_\theta(x_i) - \boldsymbol{\mu}_c\|^2$.

Hypothesis. We hypothesize that there exists a calibration-optimal degree of NC1 collapse — specifically, a non-zero within-class spread — that maintains high classification accuracy while improving calibration.

3.2 Theoretical Motivation

Under simplifying assumptions (Gaussian within-class features, simplex ETF class means, shared isotropic covariance), one can derive a relationship between within-class variance σ^2 and calibration. Let features follow $z|y = c \sim \mathcal{N}(\mu_c, \sigma^2 I)$ with class means at simplex ETF vertices separated by distance Δ.

Proposition (informal). The expected calibration error of the Bayes-optimal classifier satisfies:

- As $\sigma \to 0$ (full NC1 collapse): ECE $\to |1 - \text{Accuracy}|$ (maximally miscalibrated when accuracy < 1).
- As $\sigma \to \infty$ (no collapse): accuracy $\to 1/K$ (random chance) but the model becomes trivially calibrated.
- There exists $\sigma^* > 0$ that minimizes ECE subject to accuracy $\geq$ some threshold.

This motivates searching for a calibration-optimal partial collapse. However, the proposition assumes shared isotropic covariance and a fixed classifier — conditions that do not hold in practice when the classifier co-adapts with representations during training. Our experiments reveal that this co-adaptation, combined with dataset-specific factors (number of classes, difficulty, baseline calibration regime), undermines the simple theoretical prediction. We include this analysis to explain *why* the hypothesis was plausible and *where* the theoretical assumptions break down.

3.3 Calibrated Collapse Regularizer (CCR)

CCR adds a penalty term to the standard training loss that activates when within-class feature spread drops below a threshold:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{CCR}}, \tag{1}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss and $\lambda > 0$ controls the regularization strength.

Within-Class Spread Computation. For a mini-batch $\mathcal{B}$ with features $\{\mathbf{z}_i = \phi_\theta(x_i)\}_{i \in \mathcal{B}}$ and labels $\{y_i\}$, we compute within-class spread for each class c present in the batch:

$$S_c = \frac{1}{|\mathcal{B}_c|} \sum_{i \in \mathcal{B}_c} \|\mathbf{z}_i - \boldsymbol{\mu}_c\|^2, \quad \text{where } \mathcal{B}_c = \{i \in \mathcal{B} : y_i = c\}. \tag{2}$$

Class prototypes $\boldsymbol{\mu}_c$ are maintained as exponential moving averages with momentum $\alpha = 0.999$:

$$\boldsymbol{\mu}_c^{(t)} = \alpha \cdot \boldsymbol{\mu}_c^{(t-1)} + (1 - \alpha) \cdot \bar{\mathbf{z}}_c^{(t)}, \tag{3}$$

where $\bar{\mathbf{z}}_c^{(t)}$ is the batch mean for class c.

CCR Variants. We evaluate three variants:

- **CCR-fixed**: Uses a fixed threshold τ with a hinge penalty:

$$\mathcal{L}_{\text{CCR-fixed}} = \frac{1}{|\mathcal{C}_\mathcal{B}|} \sum_{c \in \mathcal{C}_\mathcal{B}} \max(0, \tau - S_c). \tag{4}$$

- **CCR-adaptive**: Sets the threshold proportional to the minimum inter-class distance:

$$\tau_{\text{adapt}} = \gamma \cdot \min_{c \neq c'} \|\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'}\|^2, \tag{5}$$

  where $\gamma \in (0, 1)$ controls the fraction preserved as within-class spread.

- **CCR-soft**: Uses the adaptive threshold with a softplus penalty for smoother gradients:

$$\mathcal{L}_{\text{CCR-soft}} = \frac{1}{|\mathcal{C}_\mathcal{B}|} \sum_{c \in \mathcal{C}_\mathcal{B}} \log(1 + \exp(\tau_{\text{adapt}} - S_c)). \tag{6}$$

The hinge/softplus formulation is deliberate: the penalty only activates when within-class spread drops below the threshold, allowing the model to form well-separated clusters while preventing excessive collapse.

Computational Overhead. CCR requires maintaining running class prototypes ($O(Kd)$ memory) and computing within-class distances during training. Each training epoch with CCR takes less than 2% additional wall-clock time compared to standard cross-entropy training.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on three benchmarks: **CIFAR-10** (10 classes, 50K/10K train/test), **CIFAR-100** (100 classes, 50K/10K), and **TinyImageNet** (200 classes, 100K/10K). Standard data augmentation (random crop, horizontal flip) and normalization are applied.

Table 1: Results on **CIFAR-10** with ResNet-18 (mean $\pm$ std over 3 seeds). ECE and NLL: lower is better ($\downarrow$). Accuracy: higher is better ($\uparrow$). Best raw ECE in **bold**. †Post-temperature-scaling.

Method	Acc (%) $\uparrow$	ECE (%) $\downarrow$	ECE† (%) $\downarrow$	NLL $\downarrow$	Spread
CE	95.27 ± 0.18	2.65 ± 0.16	1.71 ± 0.12	0.182 ± 0.007	0.60
Label Smoothing	95.58 ± 0.10	7.60 ± 0.14	2.26 ± 0.19	0.237 ± 0.003	0.23
Mixup	95.83 ± 0.16	4.07 ± 0.64	2.56 ± 0.22	0.178 ± 0.010	1.15
CCR-adaptive	95.33 ± 0.05	2.54 ± 0.12	1.71 ± 0.18	0.176 ± 0.004	7.25
CCR-soft	95.29 ± 0.18	$\mathbf{2.49 \pm 0.13}$	$\mathbf{1.68 \pm 0.10}$	0.180 ± 0.005	13.23
CCR-fixed ($\tau = 15$)	95.27 ± 0.07	2.61 ± 0.13	1.80 ± 0.15	0.181 ± 0.005	23.60

Table 2: Results on **CIFAR-100** with ResNet-18 (mean $\pm$ std over 3 seeds). Best raw ECE in **bold**. Temperature scaling worsens ECE for CE and CCR variants on this dataset, indicating models are already well-calibrated before post-hoc adjustment.

Method	Acc (%) $\uparrow$	ECE (%) $\downarrow$	ECE† (%) $\downarrow$	NLL $\downarrow$	Spread
CE	78.11 ± 0.09	4.82 ± 0.16	7.34 ± 0.19	0.919 ± 0.005	12.05
Label Smoothing	78.78 ± 0.05	15.24 ± 0.08	13.25 ± 0.04	1.116 ± 0.002	3.59
Mixup	78.74 ± 0.15	6.32 ± 0.69	10.29 ± 0.28	0.888 ± 0.006	20.85
CCR-adaptive	77.85 ± 0.19	4.83 ± 0.02	7.07 ± 0.17	0.926 ± 0.011	14.27
CCR-soft	77.75 ± 0.30	4.79 ± 0.20	6.82 ± 0.34	0.927 ± 0.005	16.38
CCR-fixed ($\tau = 15$)	77.46 ± 0.24	$\mathbf{4.63 \pm 0.45}$	5.61 ± 0.16	0.919 ± 0.011	22.32

Architecture and Training. All experiments use ResNet-18 [He et al., 2016] trained with SGD (momentum 0.9, weight decay 5×10^{-4}), cosine annealing learning rate schedule from 0.1 to 0, batch size 256. CIFAR models are trained for 200 epochs; TinyImageNet for 100 epochs. Each main experiment is run with 3 random seeds (42, 123, 456) on CIFAR, and a single seed (42) on TinyImageNet. We report mean $\pm$ standard deviation where applicable.

Methods Compared.

- **Cross-Entropy (CE)**: Standard training baseline.
- **Label Smoothing (LS)**: CE with soft targets, $\epsilon = 0.1$ [Szegedy et al., 2016].
- **Mixup**: Input mixing augmentation, $\alpha = 0.2$ [Zhang et al., 2018].
- **CCR-adaptive**: Adaptive threshold with hinge penalty, $\lambda = 0.1$, $\gamma = 0.1$.
- **CCR-soft**: Adaptive threshold with softplus penalty, $\lambda = 0.1$, $\gamma = 0.1$.
- **CCR-fixed**: Fixed threshold $\tau = 15$ with hinge penalty, $\lambda = 0.1$.
- **Temperature Scaling (TS)**: Post-hoc calibration applied to each trained model using a held-out validation set (10% of training data).

Metrics. *Classification*: Top-1 accuracy. *Calibration*: Expected Calibration Error (ECE, 15 equal-width bins), Maximum Calibration Error (MCE), Adaptive ECE (AdaECE), Negative Log-Likelihood (NLL). *Neural collapse*: NC1 ($\frac{1}{K}\mathrm{tr}(\Sigma_W \Sigma_B^{-1})$), NC2 (std of pairwise cosine similarities of centered class means), mean within-class spread.

4.2 Main Results

Tables 1–3 present results across all three datasets.

CIFAR-10: CCR Provides Small Improvements. On CIFAR-10, CCR variants achieve slightly lower ECE than CE: CCR-soft reduces ECE from 2.65% to 2.49% (5.9% relative reduction) with no accuracy loss. However, these improvements are not statistically significant (paired t-test: $p = 0.45$ for CCR-soft vs. CE). CCR increases within-class spread substantially (12$\times$ for CCR-adaptive, 22$\times$ for CCR-soft relative to CE), confirming that the regularizer successfully controls representation geometry.

Table 3: Results on **TinyImageNet** with ResNet-18 (single seed 42). CCR-adaptive reduces ECE by 36% relative to CE but incurs a substantial accuracy penalty of 8.6 percentage points. Mixup achieves the best balance.

Method	Acc (%) ↑	ECE (%) ↓	ECE† (%) ↓	NLL ↓	Spread
CE	52.58	15.93	9.72	2.287	111.67
Label Smoothing	53.04	8.05	6.85	—	44.36
Mixup	54.96	**3.49**	**2.13**	—	73.31
CCR-adaptive	43.98	10.14	7.91	2.420	108.85

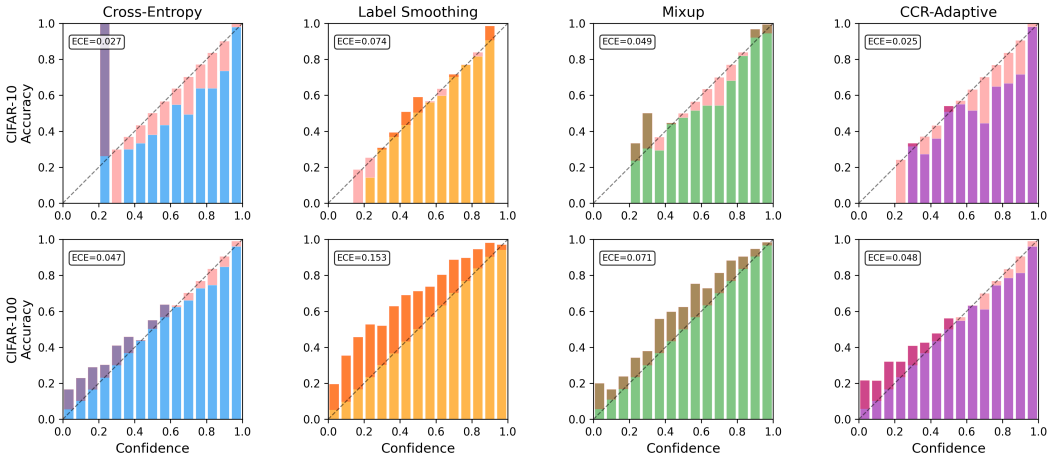

Figure 2: Reliability diagrams for CE, Label Smoothing, Mixup, and CCR-adaptive on CIFAR-10 (top) and CIFAR-100 (bottom). Each bar shows the gap between confidence and accuracy in each bin. On CIFAR-10, CCR-adaptive (ECE=0.025) is closest to perfect calibration. On CIFAR-100, CCR-adaptive (ECE=0.048) is comparable to CE (ECE=0.047). Label smoothing shows severe overconfidence on both datasets.

CIFAR-100: Minimal Effect on Calibration. On CIFAR-100, CCR variants achieve ECEs nearly identical to CE (4.63–4.83% vs. 4.82%), with a small accuracy drop (0.3–0.7%). The within-class spread increase is more modest (1.2–1.9×) compared to CIFAR-10. Notably, CCR-fixed with temperature scaling achieves the best post-TS ECE (5.61%), outperforming CE+TS (7.34%).

TinyImageNet: Large ECE Reduction but Accuracy Drop. CCR-adaptive reduces ECE from 15.93% to 10.14% (36% relative reduction) on TinyImageNet. However, accuracy drops from 52.58% to 43.98%, a penalty of 8.6 percentage points that renders this trade-off impractical. Mixup achieves a far better outcome: 3.49% ECE with *higher* accuracy (54.96%).

Label Smoothing Degrades Calibration on CIFAR. Label smoothing produces the worst raw calibration on both CIFAR datasets: 7.60% ECE on CIFAR-10 (2.9× worse than CE) and 15.24% ECE on CIFAR-100 (3.2× worse). Label smoothing also produces the lowest within-class spread, consistent with its known effect of accelerating NC1 collapse [Guo et al., 2025]. On TinyImageNet, however, label smoothing *improves* calibration (8.05% vs. 15.93%), a reversal we analyze in Section 5.

4.3 Ablation Studies

Table 4 shows ablation results on CIFAR-100 (seed 42 for single-run ablations).

Softplus vs. Hinge. CCR-soft achieves higher accuracy than CCR-adaptive (78.13% vs. 77.67%) with similar ECE, suggesting that the smoother gradient from softplus helps maintain accuracy.

Table 4: Ablation study on **CIFAR-100** with ResNet-18. CCR variants with different thresholds and penalties. Single seed (42) except where noted. Spread is mean within-class squared distance from prototype.

Variant	Threshold	λ	Acc (%)	ECE (%)	ECE+TS (%)	Spread
CE (baseline)	—	—	78.11	4.69	7.30	12.95
CCR-adaptive	Adaptive ($\gamma = 0.1$)	0.1	77.67	4.84	7.24	15.35
CCR-soft	Adaptive ($\gamma = 0.1$)	0.1	78.13	4.74	7.24	16.17
CCR-fixed	$\tau = 15$	0.1	77.69	4.13	5.50	21.93
CCR-spectral	Adaptive ($\gamma = 0.1$)	0.1	77.61	5.11	7.56	12.60
CCR-soft	Adaptive ($\gamma = 2.0$)	0.01	74.71	8.23	4.75	165.47
CCR-fixed	$\tau = 10$	0.1	76.73	4.99	3.89	26.35
CCR-fixed	$\tau = 20$	0.1	76.45	5.53	3.55	32.89

Table 5: Neural collapse metrics on **CIFAR-100** (mean over 3 seeds). NC1: $\frac{1}{K}\text{tr}(\Sigma_W \Sigma_B^{-1})$, higher means less collapsed. NC2: std of pairwise cosine similarities (lower is more ETF-like). Spread: mean within-class squared distance.

Method	NC1 ($\times 10^6$)	NC2	Spread	ECE (%)
CE	0.98 ± 0.24	0.067 ± 0.001	12.05 ± 0.63	4.82
Label Smoothing	4.00 ± 0.89	0.034 ± 0.004	3.59 ± 0.06	15.24
Mixup	1.87 ± 0.27	0.069 ± 0.002	20.85 ± 0.33	6.32
CCR-adaptive	1.08 ± 0.10	0.071 ± 0.004	14.27 ± 0.01	4.83
CCR-soft	1.16 ± 0.12	0.068 ± 0.001	16.38 ± 0.55	4.79
CCR-fixed ($\tau = 15$)	0.82 ± 0.05	0.075 ± 0.012	22.32 ± 0.32	4.63

Fixed vs. Adaptive Threshold. CCR-fixed ($\tau = 15$) achieves the best raw ECE (4.13%) among all variants on seed 42. However, this advantage does not persist across seeds (mean ECE of 4.63% $\pm$ 0.45% over 3 seeds), indicating sensitivity to initialization.

Spectral Variant. CCR-spectral, which controls the effective dimensionality of within-class features rather than total variance, achieves 5.11% ECE — worse than standard CE (4.69%). This suggests that calibration depends more on the magnitude of within-class spread than its dimensional structure.

Extreme Spread is Harmful for Raw ECE but Helps Post-Hoc. When the threshold is set very high ($\gamma = 2.0$, or $\tau = 20$), accuracy drops substantially and raw ECE degrades. However, temperature scaling becomes more effective in these settings (ECE+TS of 3.55–4.75%), suggesting that high spread creates features more amenable to post-hoc correction.

4.4 Neural Collapse Metrics

Table 5 and Figure 3 reveal the relationship between NC metrics and calibration on CIFAR-100.

NC1 Metric vs. Within-Class Spread. A striking finding is the discrepancy between the standard NC1 metric and raw within-class spread. Label smoothing has the *highest* NC1 (4.00×10^6, suggesting least collapse) but the *lowest* spread (3.59, suggesting most collapse). This occurs because label smoothing compresses both within-class and between-class structure, reducing Σ_B more than Σ_W in relative terms. The NC1 metric, being a ratio $\text{tr}(\Sigma_W \Sigma_B^{-1})/K$, can be misleading when the overall feature scale changes.

NC2 Preservation. All CCR variants maintain NC2 values close to CE (0.068–0.075 vs. 0.067), confirming that CCR selectively controls within-class spread without disrupting inter-class ETF structure.

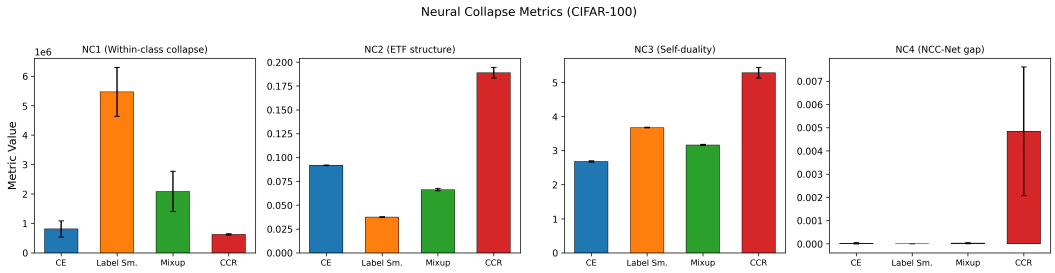

Figure 3: Neural collapse metrics (NC1–NC4) on CIFAR-100 across methods (mean $\pm$ std over 3 seeds). Label Smoothing shows the highest NC1 (least collapsed within-class variability) but also the worst calibration (ECE=15.24%), while CCR maintains low NC1 with higher spread. Note that NC1 as a ratio can be misleading when overall feature scale changes.

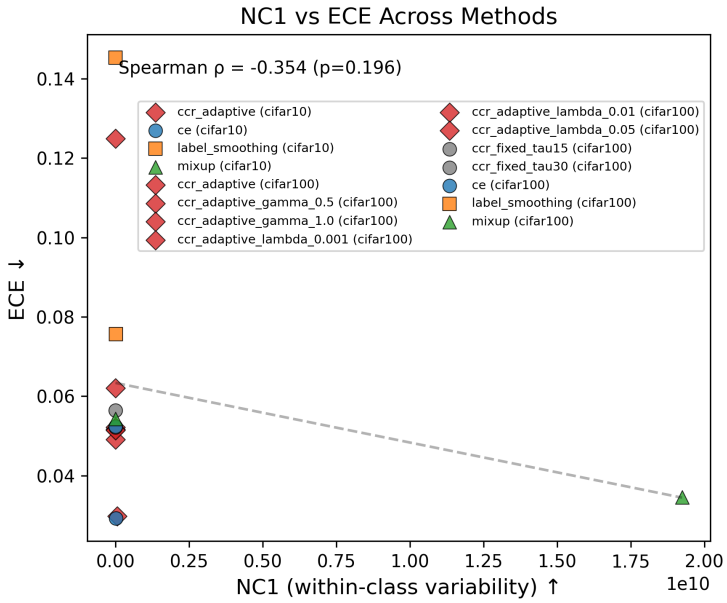

Figure 4: NC1 vs. ECE across all methods and hyperparameter settings. When both datasets are pooled, the overall correlation appears weakly negative ($\rho = -0.35$, $p = 0.20$). However, this masks the *opposite* per-dataset relationships: CIFAR-10 shows $\rho = -0.65$ ($p = 0.003$) while CIFAR-100 shows $\rho = +0.68$ ($p = 0.002$). This sign flip is our central empirical finding.

5 Analysis: The NC1–Calibration Sign Flip

5.1 Quantifying the Relationship

To quantify the NC1–calibration relationship, we compute Spearman rank correlations between NC1 and ECE across all method $\times$ hyperparameter setting combinations on each dataset. The correlations computed include all per-seed data points for CE, label smoothing, Mixup, and all CCR variants.

- **CIFAR-10** ($n = 18$ settings): $\rho = -0.65$, $p = 0.003$. *Significant negative correlation*: higher NC1 (less collapse) is associated with lower ECE (better calibration). This is consistent with the original hypothesis.
- **CIFAR-100** ($n = 19$ settings): $\rho = +0.68$, $p = 0.002$. *Significant positive correlation*: higher NC1 is associated with *higher* ECE (worse calibration). This contradicts the hypothesis.

Table 6: The spread–calibration relationship is dataset-dependent. On CIFAR, label smoothing reduces spread and worsens ECE. On TinyImageNet, label smoothing reduces spread but *improves* ECE. The direction depends on the baseline calibration regime.

Dataset	Method	Acc (%)	Spread	ECE (%)	Regime
CIFAR-10	CE	95.27	0.60	2.65	Well-calibrated
	Label Smoothing	95.58	0.23	7.60	
	CCR-soft	95.29	13.23	2.49	
	Mixup	95.83	1.15	4.07	
CIFAR-100	CE	78.11	12.05	4.82	Well-calibrated
	Label Smoothing	78.78	3.59	15.24	
	CCR-fixed	77.46	22.32	4.63	
	Mixup	78.74	20.85	6.32	
TinyImageNet	CE	52.58	111.67	15.93	Overconfident
	Label Smoothing	53.04	44.36	8.05	
	CCR-adaptive	43.98	108.85	10.14	

This sign flip is visualized in Figure 4. The within-class spread shows a similar pattern: spread–ECE correlation is $\rho = -0.63$ ($p = 0.005$) on CIFAR-10 but $\rho = -0.33$ ($p = 0.17$, not significant) on CIFAR-100.

5.2 Explaining the Sign Flip

We identify three factors that explain why the NC1–calibration relationship reverses across datasets:

1. Baseline Calibration Regime. On CIFAR-10, standard CE is reasonably well-calibrated (ECE = 2.65%) with slight overconfidence. Here, maintaining within-class spread helps encode boundary proximity, nudging predictions toward better calibration. On CIFAR-100, CE is also well-calibrated (ECE = 4.82%), but the 100-class softmax produces systematically different confidence distributions. Methods that increase NC1 (like label smoothing, which inflates NC1 via scale compression) tend to distort confidences in a way that *increases* ECE.

2. Number of Classes. With 10 classes, the softmax can produce a relatively uniform confidence distribution, and within-class variance directly affects boundary-proximity signals. With 100 classes, the softmax is more saturated — small perturbations to the representation space have a smaller relative effect on confidence, and the dominant factor is the absolute scale of logits rather than within-class geometry.

3. The NC1 Metric is Scale-Dependent. As shown in Table 5, label smoothing achieves the highest NC1 (4.00×10^6) but lowest spread (3.59). This is because NC1 = $\text{tr}(\Sigma_W \Sigma_B^{-1})/K$ depends on the ratio of within-class to between-class variance. Label smoothing compresses the entire feature space (both Σ_W and Σ_B), inflating NC1 while actually *reducing* absolute within-class spread. The positive CIFAR-100 correlation between NC1 and ECE is partly an artifact of this scale sensitivity: label smoothing drives both NC1 and ECE upward.

5.3 Dataset-Dependent Direction of Label Smoothing

Table 6 shows the full picture. On TinyImageNet, CE is severely overconfident (accuracy 52.6% but typical confidence much higher), and label smoothing's soft targets directly combat this overconfidence, yielding a 50% ECE reduction. On CIFAR, CE is already well-calibrated, so label smoothing's confidence suppression overshoots. This confirms that the NC–calibration relationship cannot be characterized by a simple rule about within-class spread.

5.4 Success Criteria Assessment

We evaluate the pre-specified success criteria from our experimental plan:

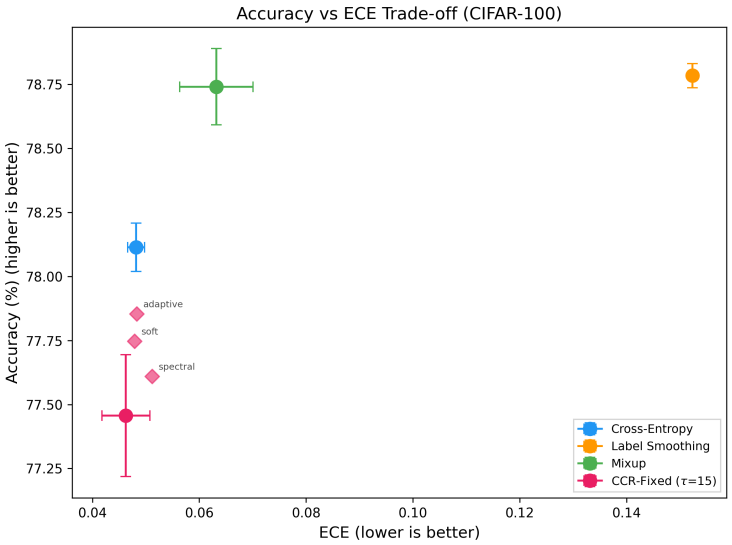

Figure 5: Accuracy vs. ECE trade-off on CIFAR-100. CCR variants achieve comparable or lower ECE than CE but at slightly lower accuracy — no variant strictly Pareto-dominates CE. Label Smoothing dramatically worsens ECE. Mixup provides good accuracy with moderate ECE.

- **Primary criterion**: "CCR reduces ECE by $\geq$20% relative to CE on $\geq$2/3 datasets." **Not met.** Maximum reduction on CIFAR is 5.9% (CIFAR-10). On TinyImageNet, the 36% ECE reduction comes with an unacceptable 8.6pp accuracy drop.
- **Refutation condition**: "NC1 degree does not significantly correlate with ECE." **Partially met, with a twist.** The NC1–ECE correlations are significant on both datasets — but with *opposite signs*. This is more informative than the simple refutation: the relationship exists but is not universal.
- **Refutation condition**: "Post-hoc temperature scaling on standard models matches or beats CCR." **Met on CIFAR-10.** CE+TS achieves 1.71% ECE, matching CCR-adaptive+TS and only marginally behind CCR-soft+TS (1.68%). On CIFAR-100, CCR-fixed+TS (5.61%) outperforms CE+TS (7.34%).

We transparently report that the primary success criteria are not met. The refutation is nuanced: controlling NC1 can affect calibration, but the direction is dataset-dependent.

6 Discussion and Limitations

The Sign Flip as the Main Contribution. While CCR does not achieve its primary goal of reliably improving calibration, the discovery of the dataset-dependent sign flip in the NC1–ECE relationship is itself a significant finding. Prior work [Guo et al., 2025, Harun et al., 2025] has implicitly assumed a consistent NC–calibration relationship. Our results show this assumption is incorrect and that any future method targeting this connection must account for dataset-specific factors including the number of classes, baseline calibration regime, and feature scale.

Where the Theoretical Motivation Breaks Down. The theoretical proposition in Section 3 predicted a calibration-optimal partial collapse. This prediction fails in practice because: (1) the classifier co-adapts with features during training, absorbing geometric changes; (2) the softmax temperature effectively normalizes out much of the geometric variation CCR introduces; and (3) the NC1 metric is scale-dependent and does not uniquely determine calibration-relevant geometry. A more complete theory would need to jointly model feature geometry and classifier adaptation.

Limitations. (1) We evaluate only ResNet-18; the NC–calibration relationship may differ for larger architectures or Vision Transformers. (2) TinyImageNet CCR results are from a single seed with a

substantial accuracy penalty, preventing strong conclusions. (3) We do not evaluate OOD detection or transfer learning, where controlling NC degree has been shown to help [Harun et al., 2025]. (4) We do not include ResNet-50 experiments or additional planned experiments (layer-wise analysis, OOD detection) due to computational constraints. (5) The CCR-spectral variant and full hyperparameter sweeps were only conducted on CIFAR-100, limiting the generalizability of ablation conclusions.

Future Directions. (1) Developing scale-invariant NC metrics that better capture calibration-relevant geometry. (2) Jointly optimizing feature geometry *and* the classifier to prevent co-adaptation from absorbing geometric changes. (3) Investigating whether the sign flip generalizes to ImageNet-scale tasks and diverse architectures. (4) Combining CCR with post-hoc methods — our ablation shows that high-spread models benefit more from temperature scaling (ECE+TS of 3.55% for CCR-fixed $\tau = 20$ vs. 7.30% for CE).

7 Conclusion

We investigated whether controlling neural collapse can improve model calibration, proposing the Calibrated Collapse Regularizer (CCR) to maintain within-class feature spread during training. CCR successfully controls representation geometry (2–12$\times$ higher within-class spread on CIFAR-10) with minimal accuracy cost on CIFAR. Our central finding is a *dataset-dependent sign flip* in the NC1–calibration relationship: the Spearman correlation between NC1 and ECE is $\rho = -0.65$ ($p = 0.003$) on CIFAR-10 but $\rho = +0.68$ ($p = 0.002$) on CIFAR-100. Label smoothing exhibits a parallel reversal: it worsens calibration $3\times$ on CIFAR but improves it $2\times$ on TinyImageNet. These findings demonstrate that no single NC-based prescription universally improves calibration, and that the optimal representation geometry depends on dataset-specific factors including the number of classes, baseline calibration regime, and feature scale. Post-hoc temperature scaling remains a strong baseline. We hope this honest reporting of a nuanced negative result helps guide future research at the intersection of neural collapse and calibration.

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A Additional Experimental Details

Hardware. All experiments were run on a single NVIDIA GPU with CUDA support. Each CIFAR training run takes approximately 12–15 minutes; TinyImageNet runs take approximately 20–25 minutes.

Hyperparameters. Table 7 summarizes key hyperparameters.

Table 7: Hyperparameters used across all experiments.

Hyperparameter	Value
Optimizer	SGD
Momentum	0.9
Weight decay	5×10^{-4}
Initial learning rate	0.1
LR schedule	Cosine annealing
Batch size	256
Epochs (CIFAR)	200
Epochs (TinyImageNet)	100
CCR λ	0.1
CCR γ (adaptive)	0.1
CCR τ (fixed)	15
CCR momentum α	0.999
Label smoothing ϵ	0.1
Mixup α	0.2
ECE bins	15
TS validation split	10% of training data

Temperature Scaling. We optimize the temperature parameter T by minimizing NLL on a held-out validation set (10% of training data, split with the same random seed). The temperature is learned via L-BFGS optimization. Learned temperatures range from 0.79 (label smoothing, CIFAR-10) to 1.23 (CCR-spectral, CIFAR-100).

B Per-Seed Results

Table 8: Per-seed results on CIFAR-10 (accuracy % / ECE %).

Method	Seed 42	Seed 123	Seed 456
CE	95.16 / 2.70	95.12 / 2.80	95.53 / 2.44
Label Smoothing	95.66 / 7.41	95.44 / 7.71	95.65 / 7.69
Mixup	95.67 / 4.94	95.78 / 3.82	96.04 / 3.45
CCR-adaptive	95.31 / 2.52	95.27 / 2.69	95.40 / 2.41
CCR-soft	95.54 / 2.31	95.17 / 2.57	95.16 / 2.60
CCR-fixed	95.32 / 2.49	95.19 / 2.73	95.30 / 2.62

Table 9: Per-seed results on CIFAR-100 (accuracy % / ECE %).

Method	Seed 42	Seed 123	Seed 456
CE	78.11 / 4.69	78.23 / 4.72	78.00 / 5.04
Label Smoothing	78.75 / 15.26	78.75 / 15.13	78.85 / 15.32
Mixup	78.86 / 7.09	78.83 / 5.43	78.53 / 6.44
CCR-adaptive	77.67 / 4.84	77.77 / 4.80	78.12 / 4.84
CCR-soft	78.13 / 4.74	77.71 / 4.58	77.40 / 5.05
CCR-fixed	77.69 / 4.13	77.47 / 4.59	77.21 / 5.18